

Enhancing DeLUCS: Improved Deep Learning-Based Unsupervised Clustering of DNA Sequences

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1 Background and Motivation

Unsupervised clustering of DNA sequences is essential in metagenomics and evolutionary biology, yet traditional alignment-based methods are computationally intensive and poorly scalable. Alignment-free alternatives, such as k-mer-based techniques, offer improved efficiency but often fail to capture long-range dependencies. Recent deep learning approaches, such as DeLUCS [1], leverage frequency chaos game representations (FCGR) to enable alignment-free clustering. However, DeLUCS relies on fully connected layers, which may limit its capacity to model complex sequence relationships.

2 DeLUCS Overview

DeLUCS is an alignment-free deep learning framework for unsupervised clustering of genomic sequences. It leverages Frequency Chaos Game Representations (FCGR) to extract features from DNA sequences and achieves improved scalability and performance compared to traditional methods. However, its reliance on fully connected layers limits its capacity to capture long-range dependencies and introduces sensitivity to k-mer size selection.

3 Problem Definition

While DeLUCS demonstrates promising results, its architectural design lacks mechanisms to effectively model long-range dependencies within sequences. This limitation can degrade clustering performance, especially for genomically diverse inputs. Furthermore, the absence of adaptive feature extraction may lead to information loss during training.

4 Proposed Solution

To overcome these limitations, we enhance DeLUCS by integrating components from modern deep learning architectures:

- **Multi-Head Self-Attention (MHSA):** Enables the model to capture long-range dependencies by attending to multiple sequence positions simultaneously [2].
- **Residual Connections:** Facilitates gradient flow and improves training stability [3].
- **Layer Normalization:** Stabilizes activations across layers, promoting faster convergence [4].
- **Hyperparameter Optimization:** We investigate configurations with 4, 8, and 16 attention heads to balance model capacity and generalization.

5 Datasets

We evaluated our model enhancements on two datasets used in the DeLUCS study:

Cypriniformes Dataset: Used in the original DeLUCS study, comprising 2,500 complete mitochondrial genomes from the Cypriniformes order, this dataset provides a biologically diverse benchmark for unsupervised clustering [5, 1]. It includes species such as carps, minnows, and loaches, representing a wide range of taxonomic diversity among freshwater vertebrates.

Bacteria Dataset: A synthetically generated metagenomic dataset designed to assess the model’s ability to cluster diverse bacterial sequences, including clinically relevant *Staphylococcus* species [9, 10, 11, 1]. The dataset contains complete genome sequences from multiple *Staphylococcus* strains, capturing genetic variability relevant to antibiotic resistance and pathogenicity.

6 Results

6.1 Evaluation Metric and Loss Function

Evaluation Metric. Clustering performance followed the accuracy metric defined in DeLUC study, computed as:

$$ACC = \frac{\sum_{i=1}^n O\{l_i = f(c_i)\}}{n}, \quad (1)$$

where n is the number of samples, l_i is the true label, c_i is the predicted cluster label, $f(c_i)$ is the optimal label mapping obtained via the Hungarian algorithm, and O is the indicator function. This metric enables a standardized comparison between predicted clusters and ground-truth taxonomic labels. All evaluations were performed on the Cypriniformes and Bacteria datasets using the same protocol as DeLUCS for fair benchmarking.

Loss Function. We employed the *Invariant Information Distillation (IID)* loss [6], which maximizes the mutual information between cluster assignments of paired augmentations, encouraging consistent and informative representations in an unsupervised setting.

6.2 Experiments

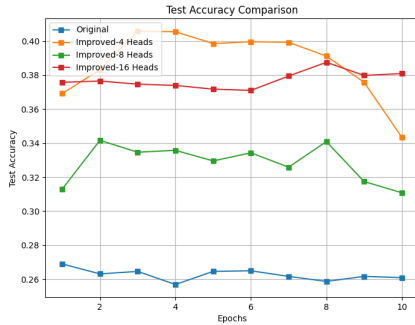
Training and Hyperparameter Setup. To evaluate our proposed model enhancements, we adopted the training procedure and evaluation methodology used in the original DeLUCS study. All models were trained for 10 epochs using a batch size of 512, with the Adam optimizer and a learning rate of 0.0001. The loss function employed was *Invariant Information Distillation (IID) Loss* [6], which maximizes the mutual information between paired samples.

Model performance was evaluated using the **Clustering Accuracy (ACC)** 1, computed via the Hungarian algorithm to determine the optimal one-to-one mapping between predicted and true labels.

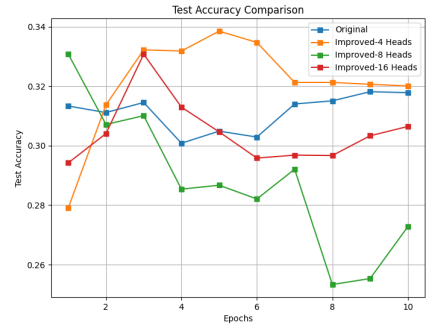
We compared the original DeLUCS architecture (which uses fully connected layers) with our improved variant incorporating Multi-Head Self-Attention (MHSA). Configurations with 4, 8, and 16 attention heads were tested to assess the trade-off between performance and model complexity. Each experiment was repeated multiple times, and results were averaged to account for variability.

6.3 Training Curve Analysis

To evaluate model performance and training stability, we tracked test accuracy and training loss over 10 epochs for each model configuration. Figures 1 and 2 present the results for both the Cypriniformes and Bacteria datasets.



(a) Test accuracy on the **Cypriniformes** dataset.

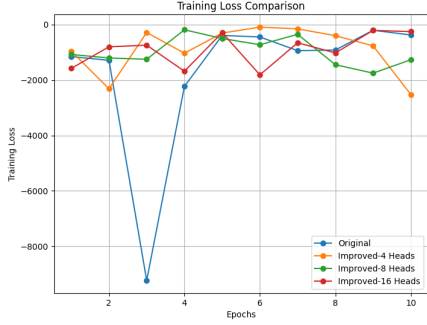


(b) Test accuracy on the **Bacteria** dataset.

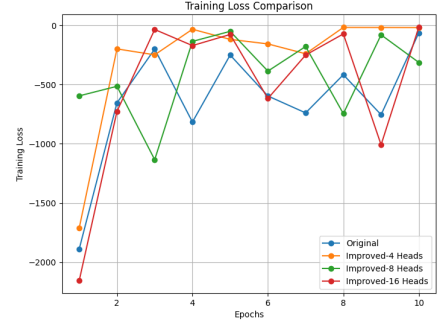
Figure 1: Test accuracy comparison across **Cypriniformes** and **Bacteria** datasets.

Performance Analysis. As illustrated in Figures 1 and 2, incorporating Multi-Head Self-Attention (MHSA) significantly improved clustering performance when using 4 attention heads. This configuration achieved the highest mean accuracy on both datasets—**45.49%** for Cypriniformes and **48.20%** for Bacteria—compared to the baseline DeLUCS (**28.75%** and **30.12%**, respectively).

In contrast, configurations with 8 and 16 heads showed reduced accuracy and more unstable training dynamics, suggesting overfitting or optimization difficulties. These findings underscore the importance of tuning attention depth to balance model capacity and generalization.



(a) Training loss on the **Cypriniformes** dataset.



(b) Training loss on the **Bacteria** dataset.

Figure 2: Training loss comparison across **Cypriniformes** and **Bacteria** datasets.

6.4 Comparison of Model Performance

Table 1 summarizes the mean test accuracy for each model configuration on the Cypriniformes and Bacteria datasets.

The results show that adding Multi-Head Self-Attention (MHSA) significantly improves clustering performance, especially with 4 attention heads. This configuration boosted accuracy from 28.75% to 45.49% on Cypriniformes and from 30.12% to 48.20% on Bacteria. However, using 8 or 16 heads led to lower accuracy, likely due to overfitting or reduced generalization.

Dataset	Model Type	Number of Heads	Mean Test Accuracy
Cypriniformes	Baseline (DeLUCS)	N/A	0.2875
	MHSA (Improved)	4	0.4549
	MHSA (Improved)	8	0.2178
	MHSA (Improved)	16	0.2027
Bacteria	Baseline (DeLUCS)	N/A	0.3012
	MHSA (Improved)	4	0.4820
	MHSA (Improved)	8	0.2751
	MHSA (Improved)	16	0.2633

Table 1: Comparison of mean test accuracy across model configurations for both **Cypriniformes** and **Bacteria** datasets.

These findings confirm that incorporating MHSA can enhance clustering performance, but only when the attention mechanism is properly configured. Future work should focus on adaptive attention strategies and regularization techniques to balance expressiveness and generalization across diverse datasets.

7 Conclusion

Our results demonstrate that incorporating Multi-Head Self-Attention (MHSA) into DeLUCS improves its ability to capture long-range dependencies in genomic sequences, leading to better clustering accuracy. The best performance was achieved with a moderate number of attention heads (e.g., 4), while higher configurations introduced instability,

likely due to overfitting or diluted representational focus.

These findings highlight the importance of balancing model complexity and generalization, especially in unsupervised settings. While MHSA brings benefits, its effectiveness depends on careful tuning and dataset characteristics. This work also emphasizes the need to evaluate deep learning innovations within domain-specific contexts rather than assuming direct transferability from fields like NLP.

Future directions include exploring adaptive attention mechanisms and extending the model to larger and more diverse genomic datasets to assess scalability and robustness in real-world bioinformatics applications.

References

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